

Dual mapping, in plain language

A short walkthrough of the 11 September 2026 freeze

Same job as `progress-notes-simple.pdf`: idea first, then click.

Map viewer: <http://127.0.0.1:8263/> Curriculum: <http://127.0.0.1:8264/>

Full record: `notes/mapping-notes.pdf` Code: `notes/mapping-code-walkthrough.pdf` Track: `notes/CXR-Mapping-Curriculum.pdf`

Read this if you want the story, then open the viewer.

Read the full notes if you want every cell count and leftover ID.

Read the code PDF if you want copy-paste Python.

Read the curriculum PDF if you want MAP-00...09 as a course.

1 The idea in one picture

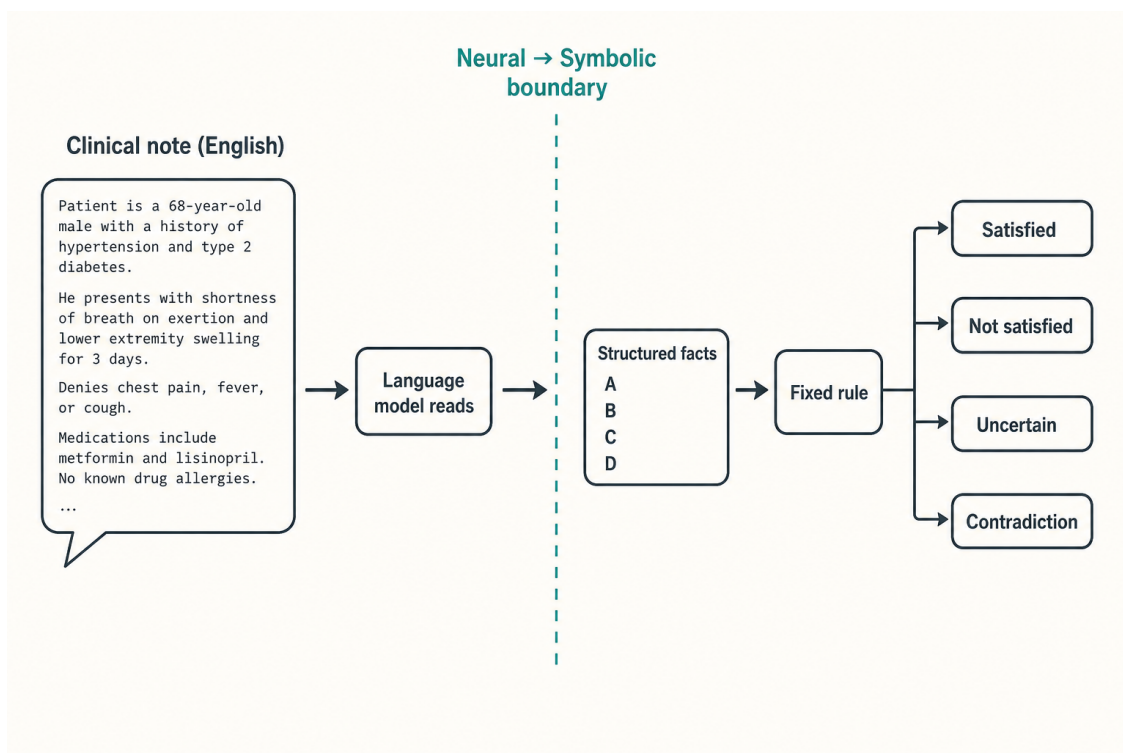


Figure 1: A language model fills an extract. A fixed rule — not the model — issues the label.

This mapping pass never rewrote that rule. We only edited the **extract** (Translate), then ran the same `ground()` and the same rule.

2 What we set out to do

Ambitious sentence (later thesis, not this freeze): automatically correct every failure mode in a large language model.

Smaller question we actually ran:

When Dual is wrong on “did first-line therapy fail?”, can we *group* those mistakes and *patch the extract* without breaking notes that were already right — and later *choose a patch*?

Answer we froze: grouping and patching — partial yes. Choosing among independent editors — no. Generic auto-correct — no.

3 The Dual question

Synthetic notes (not real patients). Four facts plus a conflict flag:

Atom	Plain meaning
A	A first-line therapy is identified
B	It was actually given, not only planned
C	Something that counts as failure happened
D	That failure is <i>of</i> first-line
X	Two claims cannot both be true

Rule (unchanged): if X then CONTRADICTION; else if any of A–D is false then NOT_SATISFIED; else if any is unknown then UNCERTAIN; else SATISFIED.

4 What we built

Nested Translate patches **none** → **G3** → **G4** → **G6** → **G7** → **G8** → **G9**. Deeper includes earlier. That is one class of edit, not six products.

On 108 Dual-fill notes: raw Dual-wrong 30; after G9 leftover 6. P-neg leak 0 after G3. Design P-bind 7/7 match after G9.

5 Walk the GUI

Start the viewer:

```
cd ~/staging/cxr-evidence-grounding-lab
./scripts/run_map_viewer.sh
```

Open <http://127.0.0.1:8263/>. Read the yellow banner (mapping, not selection).

1. Click **T1**. Gold SATISFIED. Every row already matches. This note needed no patch.
2. Click **I2**. Raw extract is NOT_SATISFIED (**admin=planned**). G9 flips planned → given. SATISFIED.
3. Click **C4**. Gold CONTRADICTION. G4 is the first match; G8 is a deeper signature.
4. Click **T2** (or filter “Still miss after G9”). Too early for a failure event. Leave it unrepaired.

Curriculum reading UI: <http://127.0.0.1:8264/>. Frozen locator :8260 is a different product — do not expand it here.

6 What we are (and are not) claiming

- **Are:** Dual-shaped extract failures can be grouped; some can be patched on the Translate side with measured collateral.
- **Are not:** a generic LLM auto-corrector; locator-chosen Encode vs Compute vs Translate; six independent editors; leftover-6 as unfinished homework.

7 Where to go next

Want...	Open...
Click notes	http://127.0.0.1:8263/
Study modules	http://127.0.0.1:8264/
Every number	notes/mapping-notes.pdf
Runnable Python	notes/mapping-code-walkthrough.pdf
Course PDF	notes/CXR-Mapping-Curriculum.pdf
Older N2S notebook	notes/progress-notes.pdf
Older simple guide	notes/progress-notes-simple.pdf
Design doc	docs/TRACKB-FAILURE-INTERVENTION-MAP.md

Rebuild: `./build-mapping-pdfs.sh` from the `notes/` directory.